

Abhishek Rout

Curriculum Vitae · October 2026

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Research Interests

Modified gravity, in particular Chern–Simons and other parity-violating extensions of general relativity; black hole and cosmological perturbation theory; gravitational-wave phenomenology; quantum field theory in curved spacetime; early-universe cosmology.

Education

Ph.D. in Physics, University of South Carolina, Columbia, SC, USA 2019–2025

Awarded 15 December 2025 · GPA 3.82/4.00

Advisor: Prof. Brett Altschul

Dissertation: *Gravitational Perturbations and Quantum Field Interactions in Modified Spacetime Backgrounds*

Integrated BS–MS in Physics, IISER Kolkata, India 2014–2019

CGPA 8.74/10 (top 10% of cohort)

Publications

Peer-Reviewed

1. A. Rout and B. Altschul, “First-Order Axial Perturbation of the Reissner–Nordström Metric Against a Possible Parity-Violating Gravity Background,” *Symmetry* **18**, 560 (2026), doi:10.3390/sym18040560, arXiv:2510.01106 [gr-qc].
2. A. Rout and B. Altschul, “Primordial Gravitational Wave Birefringence in a de Sitter Background with Chern–Simons Coupling,” *Universe* **12**, 6 (2026), doi:10.3390/universe12010006, arXiv:2512.03919 [gr-qc].
3. A. Rout and B. Altschul, “Bound States and Particle Production by Breather-Type Background Field Configurations,” *Symmetry* **16**, 1571 (2024), doi:10.3390/sym16121571, arXiv:2410.09273 [hep-th].

Submitted

4. A. Rout and B. Altschul, “Gravitational Wave Parity-Violating Strain from Pulsar Glitches in Chern–Simons Modified Gravity,” submitted to *Phys. Rev. D*, arXiv:2608.24002.
5. A. Rout, “Analytic parity-violating greybody factors of a dynamical Chern–Simons black hole,” submitted to *General Relativity and Gravitation*, arxiv:2610.10280

In Preparation

6. A. Rout and B. Altschul, “Mixed-Order Parity-Violating Corrections to Rotating Black Holes in Combined Scalar–Gauss–Bonnet and Dynamical Chern–Simons Gravity.”
7. A. Rout and B. Altschul, “Isospectrality Breaking in the Axial Quasinormal Modes of Reissner–Nordström Black Holes in Dynamical Chern–Simons Gravity.”
8. A. Rout and B. Altschul, “Rotation-Induced Isospectrality Breaking of Charged Black Holes in Dynamical Chern–Simons Gravity.”

Research Experience

Doctoral Research, University of South Carolina 2019–2025

Advisor: Prof. Brett Altschul. Parity-violating extensions of general relativity, principally Chern–Simons modified gravity: axial perturbations of Reissner–Nordström black holes; chiral corrections to tensor modes in de Sitter space and the resulting primordial gravitational-wave birefringence; and parity-violating gravitational-

wave strain from pulsar glitches. A separate project treated fermion bound states and particle production in oscillating, breather-type background fields. The work is largely analytical, supported by numerical and symbolic computation in Python and Mathematica.

Master's Thesis, IISER Kolkata

2018–2019

Supervisor: Prof. Narayan Banerjee. Thesis: Gravitational Collapse in a Spherically Symmetric Space-time in presence of a Scalar Field.

Analytical study of gravitational collapse in spherically symmetric spacetimes, including scalar-field-driven collapse with quadratic potentials in dust-dominated cosmological models.

Undergraduate Research Project, IIST Trivandrum

2017

Supervisor: Prof. Sudheesh Chhetil. Loop quantum gravity and spin-network states; applications of entangled loop quantum states to quantum-optics protocols.

Advanced Theoretical Project, IISER Kolkata

2016

Numerical study of the twin paradox in Schwarzschild spacetime (Python).

Talks and Workshops

- “Chern-Simons theory in Reissner-Nordstrom background”, contributed talk, IUCSS Workshop on Theoretical Aspects of the Standard-Model Extension and Modified Gravity, Indiana University, Bloomington, July 2024.
- “Gravitational Effects of Lorentz Violation,” Theoretical Physics Group Seminar, University of South Carolina, January 2025.
- “Topics in Beyond the Standard Model Physics,” Theoretical Physics Group Seminar, University of South Carolina, September 2024.
- “Aspects of Lorentz Violation: Cause and Effects,” Theoretical Physics Group Seminar, University of South Carolina, September 2023.
- Workshop attended (no talk): IUCSS Workshop on Gravitational Aspects of Lorentz Violation, Indiana University, Bloomington, 2023.

Teaching and Outreach

Teaching Assistant, University of South Carolina

2019–2025

- PHYS 201, General Physics I (mechanics), Semesters VIII, IX, X: led weekly discussion sections of about 40 students; wrote and graded homework, quizzes and exam problems.
- PHYS 202, General Physics II (electromagnetism), Semesters II–VIII: assisted with laboratory sessions; prepared tutorial material and Python demonstrations.

Science Outreach Volunteer, IISER Kolkata

2015–2019

Physics demonstration workshops (mechanics, wave optics, astronomy) for local high-school students.

Fellowships and Service

- INSPIRE Fellow, Department of Science and Technology, Government of India, 2014–2019.
- Co-referee, with Prof. B. Altschul, for *Phys. Rev. D*.

Computational Skills

Python, C, MATLAB, Mathematica; numerical solution of ODE and PDE systems; symbolic tensor algebra; L^AT_EX; Linux.

References

Prof. Brett Altschul

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